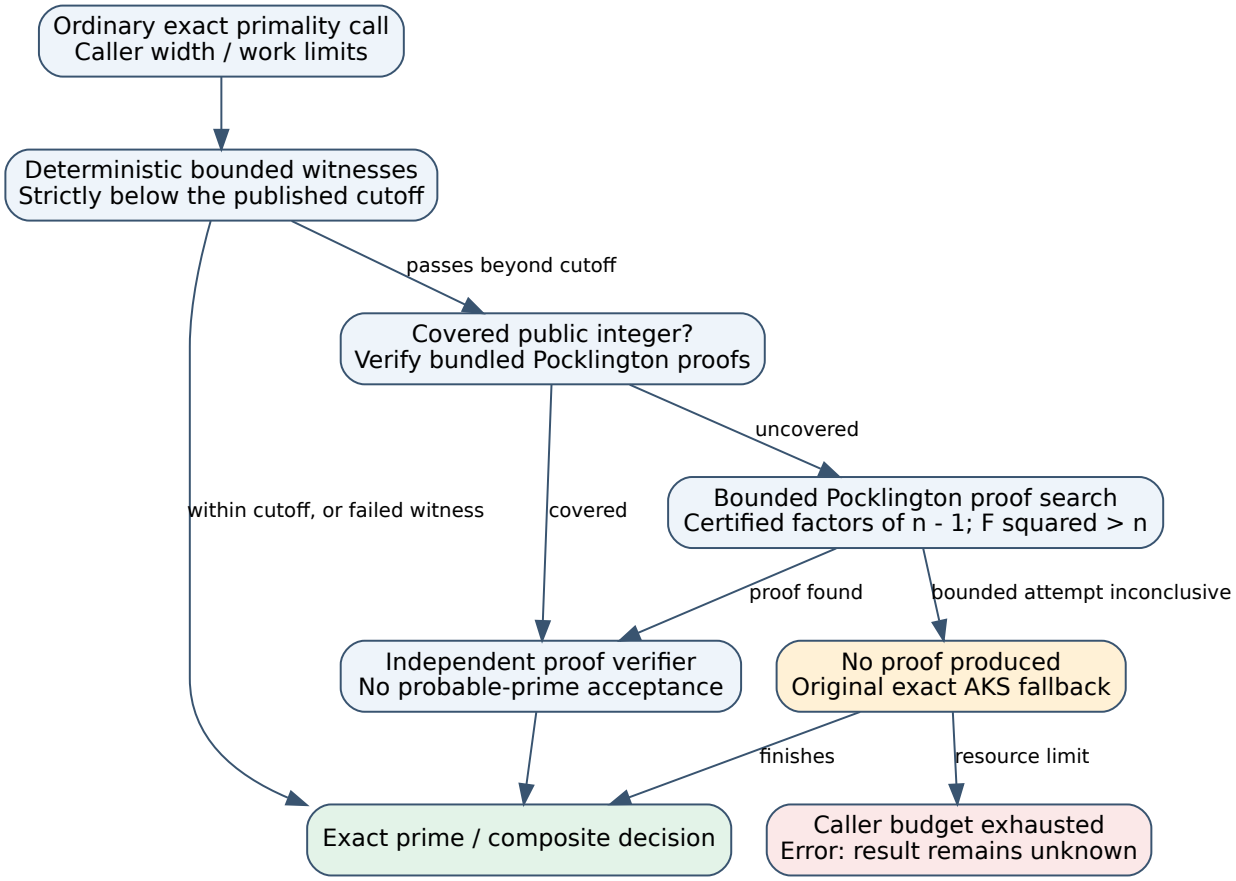
Exact primality setup: bounded tests and checked proofs

2026-10-30 · native development validation · no timing or challenge-solve claim

The ordinary >64-bit primality call previously selected AKS immediately. The implementation now uses the published thirteen-base deterministic range, automatically verifies bundled public ECCp-131 proofs, and attempts a bounded Pocklington proof for other large candidates. Successful proofs are independently replayed. AKS remains the exact fallback when no proof is produced.



Correctness boundary. The cutoff 3317044064679887385961981 is exclusive. Passing witnesses above it never certifies primality. Factoring n-1 is bounded; failure is inconclusive. Cached JSON parsing never bypasses the caller's proof verification or resource limits. The original ECCp-131 input exercises automatic verification; full relation search and the contraction crossover are unchanged.

Sources. Sorenson/Webster, arXiv:1509.00864, Theorem 1.1; Handbook of Applied Cryptography, Chapter 4 Fact 4.40 and Chapter 3 Algorithm 3.9. See README.md for links, commands, source identities, raw development records and outstanding hard-factorization / admitted-measurement requirements.